

Statistics for Land Management

Science

May, 2026

Statistics for Land Management (A07550)

Short Title: Statistics for Land Management
Faculty Science and Computing
Department Science
Cluster Mathematics and Physics
Author AHENNESSY
Credits 5

Level: Advanced

Description of Module / Aims

The module is designed to introduce key concepts in statistics for research in agriculture, forestry and horticulture. Students will learn how statistical tools can be applied by modelling relevant data on statistical software, as well as analysing and interpreting the results.

Programmes

SCIE-0069	BSc (Hons) in Land Management (in Agriculture) (WD_SBSAG_B)	1 / 1 / M
SCIE-0069	BSc (Hons) in Land Management (in Forestry) (WD_SBSFO_B)	1 / 1 / M
SCIE-0069	BSc (Hons) in Land Management (in Horticulture) (WD_SBSHO_B)	1 / 1 / M

Indicative Content

- Basic statistics: data collection methods; presentation of data; measures of center and spread
- Regression: least squares line; Pearson's correlation coefficient; the coefficient of determination; rank correlation
- Inferential statistics: sampling distributions; hypothesis testing; confidence intervals; ANOVA; Chi-squared tests

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Choose appropriate data collection methods and apply relevant graphical and analytical techniques for the investigation of data.
2. Formulate regression and correlation techniques in appropriate sets of data.
3. Set up a broad range of analytical statistical techniques and interpret the results correctly.
4. Select appropriate statistical software tools to analyse and interpret data.

Learning and Teaching Methods

- Lectures.
- Practical labs.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	75%	1,2,3
Continuous Assessment	25%	
Assignment	25%	1,2,3,4

Assessment Criteria

<40%: Very limited knowledge and understanding of statistics.

40%-49%: Demonstrate a limited knowledge of statistics.

50%-59%: Demonstrate satisfactory general knowledge of statistics.

60%-69%: Demonstrate sound knowledge of statistics. Show ability to analyse and logically argue in an effective and mature style.

70%-100%: Demonstrate authoritative handling of complex material and a highly developed and extensive understanding of statistics.

Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the final exam and continuous assessment components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Practical	12	
Independent Learning	87	

Essential Material

- "Statistics for Land Management moodle page." <https://moodle.wit.ie>

Supplementary Material

- Nageswara, R. _Statistics for Agricultural Sciences_. India: BS Publications, 2007.
- Samuels, M. _Statistics for the life sciences_. 4th ed. New Jersey: Pearson Education, 2012.

Requested Resources

- Room Type: Computer Lab